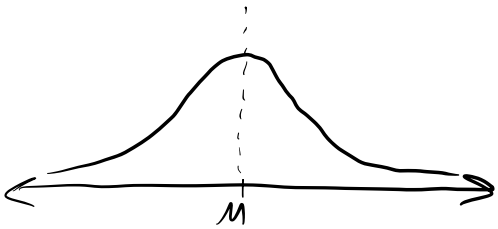
STAT 131: Lecture Notes 9

Winter 2024



8.1 The Normal Distribution

Definition 8.1. A continuous random variable X is said to follow a Normal or Gaussian distribution if X has pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \text{ for } -\infty < x < \infty,$$

where $\sigma > 0$. The parameter μ is called the mean and the parameter σ is called the standard deviation. We can write $X \sim \text{Normal}(\mu, \sigma^2)$.

Definition 8.2. A continuous random variable X is said to follow a standard Normal distribution if $X \sim \text{Normal}(0, 1)$. In other words,

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, \text{ for } -\infty < x < \infty.$$

Properties of the Normal Distribution

- The pdf has a "bell" shape
- The pdf is symmetric around μ
- $\mathbb{E}[X] = \mu$
- $V(X) = \sigma^2$

Theorem 8.1 (Moment Generating Function)**.** *If $X \sim Normal(\mu, \sigma^2)$, then the moment generating function of X is*

$$\psi_X(t) = e^{\mu t + \sigma^2 t^2/2}.$$

Example 8.1. Find the expected value and variance of the Normal distribution.

$$\psi_X'(t) = \frac{d}{dt}\psi_X(t) = e^{\mu t + \sigma^2 t^2/2} \cdot (\mu + \sigma^2 t) \Rightarrow E[X] = \psi_X'(t)\Big|^{t=0} = \boxed{\mu}$$

$$\psi_X''(t) = (\mu+\sigma^2 t)^2 e^{\mu t + \sigma^2 t^2/2} + \sigma^2 e^{\mu t + \sigma^2 t^2/2}$$

$$E[X^2] = \psi_X''(t)\Big|^{t=0} = \mu^2 + \sigma^2$$

$$V(X) = E[X^2] - (E[X])^2 = (\mu^2 + \sigma^2) - \mu^2 = \sigma^2$$

♢

The Normal distribution can be challenging to work with because the cdf is not known in closed form. We often use $\Phi(x)$ to represent the cdf of the standard normal distribution, but this must be evaluated numerically. Alternatively, tables are often used that report these probabilities.

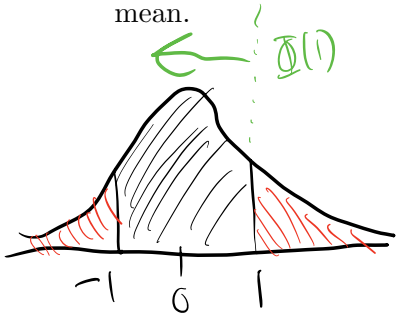
Definition 8.3. Let $X \sim \text{Normal}(0,1)$. Then, the cdf of X is denoted as

$$\Phi(x) = F(x) = \mathbb{P}(X \le x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt.$$

Example 8.2. Here is a piece of a numerical table for $\Phi(x)$. ←

x	0.5	1.0	1.5	2.0	2.5	3.0
$\Phi(x)$	0.6915	0.8413	0.9332	0.9772	0.9938	0.9987

Using this, find the probability that a Normal random variable is within one standard deviation of its mean.



$$P(-1 \le X \le 1) = P(X \le 1) - P(X \le -1)$$

$$= \Phi(1) - \Phi(-1)$$

$$= \Phi(1) - (1 - \Phi(1))$$

$$= 2\Phi(1) - 1 = \boxed{0.6826}$$

$\Phi(-1) = 1 - \Phi(1)$

♢

Example 8.3. What is the probability that a Normal random variable is within k standard deviations of the mean?

$$P(-k \le X \le k) = P(X \le k) - P(X \le -k)$$
$$= \Phi(k) - \Phi(-k)$$
$$= 2\Phi(k) - 1$$

♢

The previous example leads to the following useful facts, sometimes called the "Empirical Rule":

- The probability that a Normal random variable is within 1 standard deviation of the mean is ≈ 0.68
- The probability that a Normal random variable is within 2 standard deviations of the mean is ≈ 0.95
- The probability that a Normal random variable is within 3 standard deviations of the mean is ≈ 0.997

The empirical rule is specific to the Normal distribution. However, it is quite similar to Chebyshev's inequality which may be used to place bounds on probabilities for random variables from general distributions.

Theorem 8.2 (Chebyshev's Inequality)**.** *Let X be a random variable with finite mean $\mathbb{E}[X] = \mu$ and finite variance $V(X) = \sigma^2$. Then, for any $\epsilon > 0$,*

$$\mathbb{P}(|X - \mu| \ge \epsilon) \le \frac{\sigma^2}{\epsilon^2}$$

Example 8.4. Use Chebyshev's inequality to bound that probability that a random variable is within 1, 2, and 3 standard deviations of its mean.

1) $P(|X-\mu| \ge \sigma) \le \frac{\sigma^2}{\sigma^2} \Rightarrow P(|X-\mu| < \sigma) \ge 1 - 1 = 0$

2) $P(|X-\mu| \ge 2\sigma) \le \frac{\sigma^2}{2^2\sigma^2} = \frac{1}{4} \Rightarrow P(|X-\mu| < 2\sigma) \ge 1 - \frac{1}{4} = \frac{3}{4}$

3) $P(|X-\mu| \ge 3\sigma) \le \frac{\sigma^2}{3^2\sigma^2} = \frac{1}{9} \Rightarrow P(|X-\mu| < 3\sigma) \ge 1 - \frac{1}{9} = \frac{8}{9}$

♢

Theorem 8.3 (Converting from Normal to Standard Normal)**.** *Let $X \sim Normal(\mu, \sigma^2)$. Then $Z = \frac{X-\mu}{\sigma}$ will follow the standard Normal distribution (i.e., $Z \sim Normal(0,1)$).*

Example 8.5. Suppose that X follows a Normal distribution with mean 5 and standard deviation 2. Find $\mathbb{P}(1 \leq X \leq 8)$.

$$P(1 \leq X \leq 8) = P(1-5 \leq X-5 \leq 8-5) = P\left(\frac{1-5}{2} \leq \frac{X-5}{2} \leq \frac{8-5}{2}\right)$$

$$= P\left(-2 \leq Z \leq 3/2\right) = \Phi(3/2) - \Phi(-2)$$

♢

A very important property of the Normal distribution is the linear transformation property. This says that if X is a Normal random variable and you take a linear transformation of X, then this too will be a Normal random variable. This is not true for most other distributions (e.g., the Exponential).

Theorem 8.4 (Linear Transformation Property)**.** *Let $X \sim Normal(\mu, \sigma^2)$ and let*

$$Y = aX + b,$$

for $a > 0$. Then, $Y \sim Normal(a\mu + b, a^2\sigma^2)$.

Proof. Use Thm. 6.8 $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$ $e^{-\frac{1}{2}\left(\frac{y-(b+a\mu)}{a\sigma}\right)^2}$

$= \frac{1}{a}\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\left(\frac{y-b}{a}\right)-\mu}{\sigma}\right)^2} = \frac{1}{a\sigma\sqrt{2\pi}} e$

This is the normal pdf with mean $a\mu+b$ and variance $a^2\sigma^2$

□

Example 8.6. Suppose X represents the distance in feet that a long jumper jumps at a track meet. Also suppose that $X \sim \text{Normal}(16, 2)$. If distance were measured in inches instead of feet, what distribution would the jumper's distance have?

$Y = 12X$ $\quad E[Y] = E[12X] = 12E[X] = 12 \cdot 16 = 192$

$V(Y) = V(12X) = 12^2 V(X) = 144 \cdot 2 = 288$

$\Rightarrow Y \sim \text{Normal}(192, 288)$

♢

Theorem 8.5 (Sum of Independent Normals)**.** *Let* $X \sim Normal(\mu_x, \sigma_x^2)$ *and* $Y \sim Normal(\mu_y, \sigma_y^2)$. *Assume that* $X \perp Y$. *If* $W = X + Y$, *then*

$$W \sim Normal(\mu_x + \mu_y,\ \sigma_x^2 + \sigma_y^2).$$

Example 8.7. Let X_i denote the weight of a randomly selected prepackaged one-pound bag of carrots. Of course, one-pound bags of carrots won't weigh exactly one pound. Instead, the weights of "one-pound" bag of carrots are independently Normally distributed with a mean of 1.18 pounds and a standard deviation of 0.07 pounds. Suppose you buy three such bags of carrots. What is the distribution of the weight of all your carrots?

$X_1, X_2, X_3 \overset{iid}{\sim} \text{Normal}(1.18, 0.07^2)$ and $T = X_1 + X_2 + X_3$

$\Rightarrow T \sim \text{Normal}(3 \cdot 1.18, 3 \cdot 0.07^2)$

$\Rightarrow T \sim \text{Normal}(3.54, 0.0147)$

♢

8.2 Properties of the Sample Mean

Consider sampling the heights of 1,000 random individuals in the US and taking the average. Intuitively, this should give us a reasonable estimate of the average height of all people in the US. However, this average is a random variable. If we were to repeat our random sampling, we would get a different average each time. What can be said about the quality of this sample average? This is what we will try to answer.

Suppose $X_1, X_2, \ldots, X_n$ are independent and identically distributed random variables (we call this a random sample) that have some mean μ and some variance σ^2. We are not assuming that these are Normally distributed. We are not even assuming that they are continuous random variables...they could be discrete.

Definition 8.4 (Sample Mean)**.** The sample mean of n random variables $X_1, X_2, \ldots, X_n$ is their average:

$$\bar{X}_n = \frac{1}{n}(X_1 + X_2 + \cdots + X_n).$$

Theorem 8.6 (Mean and Variance of the Sample Mean)**.** *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Let $\bar{X}_n$ be the sample mean. Then $\mathbb{E}[\bar{X}_n] = \mu$ and $V(\bar{x}_n) = \sigma^2/n$.*

Proof. Use independent sums of variances.

$$E[\bar{X}_n] = E\left[\frac{1}{n}(X_1 + X_2 + \cdots + X_n)\right] = \frac{1}{n}\left[E[X_1] + E[X_2] + \cdots + E[X_n]\right]$$
$$= \frac{1}{n} \cdot n \cdot \mu = \mu$$

$$V(\bar{X}_n) = V\left(\frac{1}{n}(X_1 + X_2 + \cdots + X_n)\right) = \frac{1}{n^2}\left[V(X_1) + V(X_2) + \cdots + V(X_n)\right]$$
$$= \frac{1}{n^2} \cdot n \cdot \sigma^2 = \sigma^2/n$$

□

Notice that the sample mean depends on the sample size n. In other words, as we collect more data and n grows, the sample mean changes. In this way, we can think of $\bar{X}_n$ as a sequence of random variables (similar to a sequence of numbers in calculus, but random now). We might want to think about what happens for this sequence as $n \to \infty$.

Definition 8.5 (Convergence in Probability)**.** An infinite sequence of random variables $X_1, X_2, X_3, \ldots$ is said to converge to b in probability if for every number $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - b| < \epsilon) = 1.$$

This property can be expressed as $X_n \xrightarrow{p} b$, and we say that X_n converges to b in probability.

Theorem 8.7 (Law of Large Numbers)**.** *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n$ be the sample mean. Then,*

$$\bar{X}_n \xrightarrow{p} \mu.$$

In other words, the sample mean converges in probability to the true distribution mean.

Proof. Use Chebyshev's Inequality.

$$V(\bar{X}_n) = \sigma^2/n \qquad E[\bar{X}_n] = \mu$$

$$\text{so } P(|\bar{X}_n - \mu| < \varepsilon) \geq 1 - \frac{(\sigma^2/n)}{\varepsilon^2} = 1 - \frac{\sigma^2}{n\varepsilon^2}$$

$$\Rightarrow \lim_{n\to\infty} P(|\bar{X}_n - \mu| < \varepsilon) \geq 1$$

□

Theorem 8.8 (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n$ be the sample mean. Then,*

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \Phi(x).$$

In other words, if we take a random sample from any population (distribution) with mean μ and variance σ^2, then as n becomes large, the sample mean, $\bar{X}_n$ will be approximately Normal distributed with mean μ and variance $(\sigma/\sqrt{n})^2$. Similarly, the sum $\sum_{i=1}^{n} X_i = n \cdot \bar{X}_n$ will also have an approximate Normal distribution with mean $n\mu$ and variance $n\sigma^2$. Notice that it doesn't matter whether or not the original distribution we sample from is discrete or continuous!

Example 8.8 (Tossing a Coin)**.** Suppose a fair coin is tossed 900 times. Use the central limit theorem to approximate the probability of obtaining more than 495 heads.

$X_i \overset{iid}{\sim} \text{Bernoulli}(1/2)$ for $i = 1, \ldots, 900$ $\quad E[X_i] = 1/2 \quad V(X_i) = 1/4$

Let $S = \sum_{i=1}^{900} X_i$ $\quad E[S] = 900\,E[X_i] = 450$, $\quad V(S) = 900 \cdot 1/4 = 225$

$\sigma_S = \sqrt{225} = 15$

$$P(S > 495) = P\left(\frac{S - 450}{15} > \frac{495 - 450}{15}\right)$$

$$\approx P(Z > 3) = 1 - P(Z \leq 3) = 1 - \Phi(3) = 0.0013$$

♢

Example 8.9 (Sampling from a Uniform Distribution)**.** Suppose that a random sample of size $n = 12$ is taken from a uniform distribution on the interval $[0, 1]$. Use the central limit theorem to approximate the value of $\mathbb{P}\left(|\bar{X}_n - \frac{1}{2}| \leq 0.1\right)$.

$E[X_i] = 1/2 \quad V(X_i) = 1/12$

$\bar{X}_n$ is approx $\text{Normal}\left(1/2, \frac{1}{144}\right)$ ←

$$P(|\bar{X}_n - 1/2| \leq 0.1) = P(-0.1 \leq \bar{X}_n - 1/2 \leq 0.1)$$

$$= P\left(\frac{-0.1}{1/12} \leq \frac{\bar{X}_n - 1/2}{1/12} \leq \frac{0.1}{1/12}\right) = P(-1.2 \leq Z \leq 1.2)$$

$$= 2\Phi(1.2) - 1 = 0.7698$$

♢

Example 8.10 (Signal and Noise)**.** Imagine that we are trying to transmit a signal. During the transmission there are 100 sources that independently add a little bit of noise to the signal. Each noise source produces an amount of noise that is uniformly distributed between $a = -1$ and $b = 1$. If the total amount of noise is greater than 10 or less than -10, then the entire signal will be corrupted. However, if the absolute value of the total amount of noise is less then 10, then the signal will be transmitted without issue. What is the approximate probability that the signal will be transmitted without an issue?

$X_i \overset{iid}{\sim} \text{Uniform}(-1, 1) \qquad E[X_i] = 0 \quad V(X_i) = \frac{2^2}{12} = 1/3$

$S = \sum_{i=1}^{100} X_i \qquad E[S] = 100 \cdot 0 = 0 \qquad V(S) = 100 \cdot 1/3 = 100/3$

$\Rightarrow S$ is approx. $\text{Normal}(0, 100/3)$

we want $P(-10 \le S \le 10) = P\left(\frac{-10}{\sqrt{100/3}} \le \frac{S-0}{\sqrt{100/3}} \le \frac{10}{\sqrt{100/3}}\right)$

$= P(-\sqrt{3} \le Z \le \sqrt{3})$

$= 2\Phi(\sqrt{3}) - 1 \approx 0.91$

♢